

Title: A JWST Investigation of Solar Systems in Formation: Do MHD Winds Drive Protoplanetary Disk Accretion and Evolution?

Abstract: Planetary systems form within protoplanetary disks whose mass and angular momentum evolution determine the architectures of emerging planetary systems. Radially extended magnetohydrodynamic (MHD) disk winds are predicted to remove angular momentum and enable accretion, thereby regulating both the planet formation mass reservoir and radial mass distribution. However, the lack of spatially and spectrally resolved observations of disk winds means that predictions of wind mass loss rates remain untested. We propose JWST NIRSpec IFU and MIRI MRS observations of moderately inclined ($i \approx 45\text{--}75^\circ$) Class II disks to simultaneously measure wind morphology and kinematics. By deriving mass loss rates across emission lines corresponding to different originating disk radii and comparing them to known accretion rates, we will determine the radial profile of wind-driven mass removal. This program will be the first ever direct observational test of whether MHD winds can drive disk accretion. The resulting constraints on angular momentum transport will advance models of disk evolution and establish a physical framework for interpreting exoplanet demographics.

■ Scientific Justification

Disk Evolution Requires Efficient Angular Momentum Removal

Protoplanetary disks contain far more angular momentum than can be incorporated into the central star or resulting planetary system, requiring an efficient mechanism for angular momentum transport [1]. Disk material loses angular momentum as it moves inward, meaning that accretion requires angular momentum in the disk to be redistributed or removed. Simulations indicate that radially extended magnetohydrodynamic (MHD) disk winds can extract angular momentum along magnetic field lines anchored in the disk, enabling accretion while reshaping disk structure via mass removal [2].

However, whether MHD winds drive accretion remains uncertain, as their mass loss rates have not been observationally constrained. The spatial structure of winds from four edge-on disks has already been revealed by JWST, but these observations lack the velocity information required to determine their wind mass loss rates [5]. MHD winds are expected to strongly shape disk surface density, solid transport, and planetary migration. Quantifying their impact on disk structure is therefore essential for interpreting the rapidly expanding population of observed exoplanets.

We propose JWST NIRSpec and MIRI observations of protoplanetary disks spanning a range of accretion rates to measure wind morphologies and kinematics. JWST is currently the only facility capable of performing spatially and spectrally resolved observations of disk winds in the infrared, where a plethora of wind emission features lie. The spatial information from these observations combined with velocity estimates from their spectra will allow for robust derivations of the wind mass loss rates. These measurements will provide the first empirical constraints on the role of MHD winds in disk evolution and deliver critical inputs for models of star and planet formation.

Observational Confirmation of MHD Winds Driving Accretion is Missing

Simulations predict that MHD winds must remove mass at rates comparable to stellar accretion rates to extract sufficient angular momentum [5]. However, observations of mass loss rates currently only provide spatial information; the mass loss rates remain unknown. If wind mass loss rates are indeed on par with stellar accretion rates, winds would fundamentally change disk structure, controlling the distribution of solids and ultimately determining where and how planets form [2].

To obtain mass loss rates from wind observations, both the spatial structure of the wind and its velocity must be known. This requires observations of disks with inclinations small enough to discern nested wind structure, but large enough to constrain velocities from Doppler shift in their spectra. This program fills that gap by obtaining spatially and spectrally resolved observations of winds from inclined disks across a range of accretion rates. By generating the first measurements of wind mass loss rates and their scaling with accretion, we will directly test whether MHD winds are responsible for angular momentum transport in protoplanetary disks.

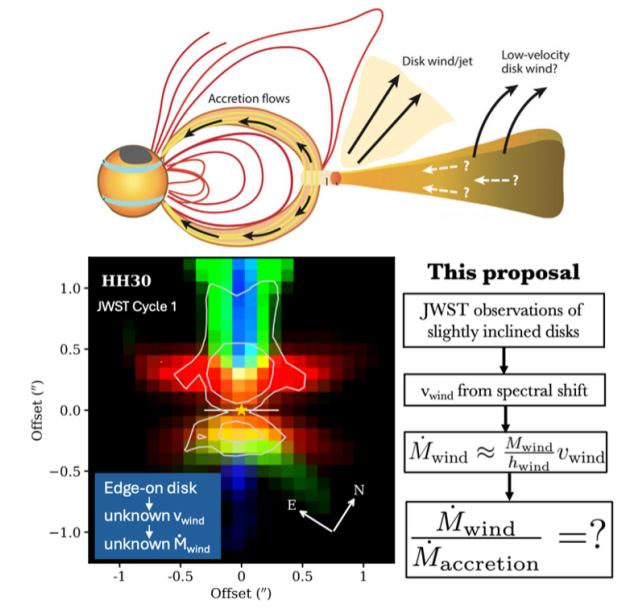


Figure 1: **This program aims to capture wind features seen in previous observations of edge-on disks while obtaining velocity information to constrain wind mass loss rates.** (Top) A schematic detailing the morphology of disk accretion and high-velocity and low-velocity wind components from [3]. (Bottom) JWST NIRSpec and ALMA observations of an edge-on disk from [6], revealing the nested chemical morphology of disk outflows. ALMA CO detections are in red, JWST H₂ detections are in green, and jet tracers are in blue. Since only edge-on disks were surveyed to discern this nested morphology, wind velocities cannot be inferred from Doppler shift in the spectra, preventing a robust estimation of the wind mass loss rates. We propose for JWST NIRSpec IFU and MIRI MRS observations of four slightly inclined ($i \approx 45 - 75^\circ$) protoplanetary disks in order to compute robust estimates of the wind mass loss rate. This program will reveal whether MHD winds drive protoplanetary disk accretion and consequently change disk evolution.

This Proposal: Observations of Inclined Disks to Derive Mass Loss Rates

Imaging and spectra of Class II (gas-rich) disks with slight inclinations ($i \approx 45 - 75^\circ$) are necessary to determine whether MHD winds drive accretion. JWST is the only facility capable of resolving both the chemical structure and kinematics of disk winds at infrared wavelengths, where key emission lines originate.

We propose NIRSpec IFU and MIRI MRS observations of ten slightly inclined ($i \approx 45 - 75^\circ$) Class II disks with known wind tracers detected by other facilities (i.e. HST, ALMA). We aim to retrieve spatially resolved emission maps of winds across the infrared, which will allow us to discern the nested chemical structure and determine the velocity from Doppler shift in the spectra. Figure 1 shows JWST Cycle 1 NIRSpec observations [6] of winds from an edge-on disk whose velocity, and thus mass loss rate, is unknown and a flow chart establishing how observations of slightly inclined disks will allow us to retrieve the necessary velocity measurements to constrain wind mass loss rates.

Our observations will produce spatial and velocity constraints on each disk wind, allowing for an estimate of mass loss rate as a function of radius and disk properties. This will allow us to determine how wind mass loss rates scale with stellar accretion rates, thus discerning whether winds remove sufficient angular momentum to drive accretion.

Disk Wind Diagnostics Are Essential to Disk Evolution and Planet Formation

The proposed measurements of spatially and spectrally resolved wind mass loss rates will provide the first direct constraints on the efficiency of MHD winds in removing angular momentum from protoplanetary disks. By establishing whether wind mass loss rates scale with accretion rates, this program will determine whether MHD winds drive disk evolution, advancing a fundamental and untested component of planet formation theory.

This program is timely because JWST has only recently revealed the spatial structure of disk winds, leaving their dynamical role unconstrained. The proposed observations directly leverage JWST’s unique capabilities and align with its “Star Lifecycle” science theme by probing the physical processes governing disk evolution and planet formation. These results will enhance the legacy value of JWST by providing benchmark datasets for disk evolution models, enabling studies of disk chemistry and jet physics, and creating community resources that inform the interpretation of exoplanet populations across multiple subfields.

References

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